

Assignment 2 – GPT-2 Text Generation Parameter Experiments

Overview

This assignment explores **Generative AI text generation using the GPT-2 language model**. The main objective is to understand how different text-generation parameters affect the output produced by GPT-2.

The experiments were implemented using **Python and Google Colab** with the Hugging Face Transformers library.

Objectives

- Load and use the GPT-2 language model.
- Generate text using a predefined prompt.
- Experiment with different text-generation parameters.
- Compare the effect of Temperature, Top-K, Top-P, and Repetition Penalty.
- Generate multiple versions of a custom prompt.
- Observe how different parameter values influence generated text.

Tools & Technologies

- Python
- Google Colab
- PyTorch
- Hugging Face Transformers
- GPT-2

Model Used

GPT-2

Model: `openai-community/gpt2`

The GPT-2 tokenizer and language model are loaded using the Hugging Face Transformers library.

Original Prompt

The main prompt used for the experiments was:

"The future of artificial intelligence will"

The model generated different continuations of this prompt by changing individual generation parameters.

Experiments

Experiment 1 – Temperature

Temperature: 0.3

A lower temperature was used to observe more focused and less varied text generation.

Experiment 2 – Temperature

Temperature: 1.5

A higher temperature was used to observe more varied and creative text generation.

Experiment 3 – Top-K

Top-K: 20

Top-K sampling was used to restrict token selection to the top 20 probable tokens.

Experiment 4 – Top-P

Top-P: 0.7

Top-P sampling was used to control token selection based on cumulative probability.

Experiment 5 – Repetition Penalty

Repetition Penalty: 1.5

A repetition penalty was applied to reduce repeated words and phrases in the generated text.

Generation Settings

The text generation function uses the following parameters:

Parameter	Purpose
<code>temperature</code>	Controls randomness in generated text
<code>top_k</code>	Limits token selection to the top K candidates
<code>top_p</code>	Controls token selection using cumulative probability
<code>repetition_penalty</code>	Helps reduce repeated text
<code>max_new_tokens</code>	Controls the maximum number of newly generated tokens

The notebook uses **80 maximum new tokens** for generation.

Custom Prompt Experiment

A custom story prompt was also tested:

"When Maya opened the old wooden box, she discovered a small device that could predict events twenty-four hours before they happened."

Three different versions were generated using different combinations of Temperature, Top-K, Top-P, and Repetition Penalty.

Custom Version 1

- Temperature: **0.5**
- Top-K: **30**
- Top-P: **0.9**
- Repetition Penalty: **1.1**

Custom Version 2

- Temperature: **0.9**
- Top-K: **50**
- Top-P: **0.95**
- Repetition Penalty: **1.1**

Custom Version 3

- Temperature: **1.2**
- Top-K: **50**
- Top-P: **0.95**
- Repetition Penalty: **1.2**

High Temperature Experiment

An additional experiment was performed using:

Temperature: 2.0

This experiment was used to observe the effect of a very high temperature on GPT-2's generated output.

Results

The experiments produced different text outputs depending on the generation parameters.

The assignment demonstrates that changing parameters such as **temperature, Top-K, Top-P, and repetition penalty** can significantly influence the style, randomness, diversity, and repetition of generated text.

Files Included

```
Assignment-2/  
├──  
├── Gen_Ai_assignment_2.ipynb  
└── README.md
```

Learning Outcomes

After completing this assignment, I learned:

- How to load and use GPT-2 for text generation.
- How text-generation parameters influence AI-generated content.
- How Temperature affects randomness.
- How Top-K controls the number of candidate tokens.
- How Top-P controls token selection based on cumulative probability.
- How Repetition Penalty can reduce repetitive outputs.
- How to compare different generated outputs.
- How to perform Generative AI experiments using Python and Google Colab.

Conclusion

This assignment provided practical experience with **GPT-2 and Generative AI text generation**. By changing Temperature, Top-K, Top-P, and Repetition Penalty, different characteristics of the generated text could be observed. The custom prompt and high-temperature experiments further demonstrated how generation settings can affect the final output.

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